

An Exact Irrational–Rational Reeb Orbit Atlas for Ellipsoids

HCS Research Program

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Abstract

We freeze the standard contact form on the boundary of the four-dimensional ellipsoid $E(a, b) = \{\pi|z_1|^2/a + \pi|z_2|^2/b \leq 1\}$. An elementary closure argument gives exactly two simple coordinate Reeb orbits when a/b is irrational. Their actions and periods, transverse return multipliers, and Conley–Zehnder indices are recorded for twelve iterates, with every $\sqrt{2}$ floor certified by integer-square inequalities. For rational $a/b = p/q$ we instead expose the full-boundary Morse–Bott family and leave the pre-perturbation CZ index undefined. This is a source-local analytic A1 theorem; it has no arithmetic owner or target determinant.

1 Frozen contact model

Let

$$E(a, b) = \left\{ (z_1, z_2) \in \mathbb{C}^2 : \frac{\pi|z_1|^2}{a} + \frac{\pi|z_2|^2}{b} \leq 1 \right\}, \quad a, b > 0,$$

and let $\lambda_0 = \frac{1}{2} \sum_j (x_j dy_j - y_j dx_j)$ on $\partial E(a, b)$. The Reeb flow and coordinate circles are

$$\varphi_t(z_1, z_2) = (e^{2\pi i t/a} z_1, e^{2\pi i t/b} z_2), \quad \gamma_1 = \{z_2 = 0\}, \quad \gamma_2 = \{z_1 = 0\}.$$

The actions equal the periods: $A(\gamma_1) = T(\gamma_1) = a$ and $A(\gamma_2) = T(\gamma_2) = b$. We use the coordinate complex-line trivialization used by Hutchings ($\xi|_{\gamma_1}$ is the second $\mathbb{C}$ summand and $\xi|_{\gamma_2}$ the first), rather than silently identifying it with a filling-disk framing.

Theorem 1 (Irrational orbit classification and index). *If $a/b \notin \mathbb{Q}$, the only simple closed Reeb orbits are γ_1 and γ_2 . For every $k \geq 1$,*

$$\begin{aligned} A(\gamma_1^k) &= T(\gamma_1^k) = ka, & \rho_\perp(\gamma_1^k) &= e^{\pm 2\pi i ka/b}, & \mu_{\text{CZ}}(\gamma_1^k) &= 2\lfloor ka/b \rfloor + 1, \\ A(\gamma_2^k) &= T(\gamma_2^k) = kb, & \rho_\perp(\gamma_2^k) &= e^{\pm 2\pi i kb/a}, & \mu_{\text{CZ}}(\gamma_2^k) &= 2\lfloor kb/a \rfloor + 1. \end{aligned}$$

All these returns are nondegenerate.

Proof. If both coordinates of a point are nonzero and it returns at time $t > 0$, then t/a and t/b are integers, forcing $a/b \in \mathbb{Q}$. Thus an irrational slope leaves only the two axes. Along γ_1^k , the transverse complex line rotates by $2\pi ka/b$, and along γ_2^k by $2\pi kb/a$. The rotation-index convention in Hutchings gives the displayed odd floor formulas; irrationality excludes an integer crossing, hence nondegeneracy. Integrating λ_0 over each coordinate circle gives its action, and the flow formula gives its period. $\square$

2 The rational Morse–Bott boundary

Suppose $a/b = p/q$ in lowest terms. The common period is $L = qa = pb$. Every point of the boundary returns at L , so the Morse–Bott critical manifold is the full three-dimensional boundary and its orbit-space family has dimension two. The coordinate circles are degenerate members with transverse multiplier one. Consequently a nondegenerate CZ integer is not assigned before a perturbation; applying an irrational floor formula here would be a category error. The receipt contains rational controls $2/1$, $3/2$, and $5/3$, including the exact resonance equation.

3 Integer-square certificates

For $a/b = \sqrt{2}$, the index of γ_1^k uses the integer m with

$$m^2 \leq 2k^2 < (m+1)^2,$$

while the swapped transverse ratio $1/\sqrt{2}$ uses

$$2m^2 \leq k^2 < 2(m+1)^2.$$

These inequalities are evaluated with integer arithmetic for $1 \leq k \leq 12$ and therefore certify every floor without a floating-point boundary test. A 90-digit calculation is used only to print the real and imaginary components of the unit multiplier; the independent checker verifies those displays.

Table 1: Finite receipt dimensions and locked route disposition.

regime	rows	exact content
irrational $\sqrt{2}$ and reciprocal	48	action, period, multiplier, CZ, square proof
rational Morse–Bott controls	6	resonance, common period, unit return, null CZ

4 Independent audit

The producer is deterministic and emits the canonical JSON receipt. A producer-independent checker makes 2,000-plus exact assertions, a SymPy program checks the flow, rotation matrix, resonance, and square identities, and byte replay compares two fresh producer runs. A hostile suite rejects 29/29 repaired-hash mutations, including altered floors, multipliers, resonance rows, scope flags, and route verdicts.

References

- [1] M. Hutchings, *Lecture notes on embedded contact homology*, arXiv:1303.5789, <https://arxiv.org/abs/1303.5789>.
- [2] H. Hofer, K. Wysocki and E. Zehnder, The dynamics on three-dimensional strictly convex energy surfaces, *Ann. of Math.* (2) 148 (1998), DOI: 10.2307/120994, <https://doi.org/10.2307/120994>.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, determinant-matching, or Hilbert–Pólya claim. **Data and code.** The receipt and audits accompany HCS-C242. This is not external peer review. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checks the displayed formulas and metadata.